PERIPHERALS AND POWER SUPPLY

Day 4: Connecting and Testing Peripherals

Practical (1 Hour): Set up and test peripherals like printers and webcams

Task: Resolve connectivity issues for a given setup.

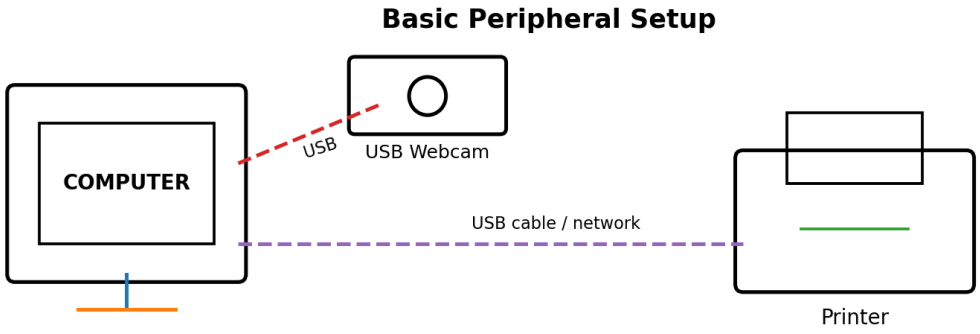


Figure: A basic computer setup with a USB webcam and printer.

Task 1: Connect a USB webcam and test it

Steps:

1. Connect the USB webcam to a USB port on the computer.
2. Wait for Windows to detect the webcam. If Windows asks for a driver, allow it to install. If needed, download the correct driver from the webcam maker's official website.
3. Open the Start menu and open the Camera app.
4. If the app asks for permission to use the camera, select Allow.
5. Check that the webcam video appears clearly.
6. Click the photo button to capture a picture.

Result: The webcam is connected and tested if the video appears in the Camera app and a photo can be captured.

Task 2: Set up a local printer and print a test page

Steps:

7. Turn on the printer and check that it has paper and enough ink or toner.
8. For a USB printer, connect its USB cable to the computer. For a network printer, connect it to the same Wi-Fi or network as the computer.
9. On Windows, open Settings > Bluetooth & devices > Printers & scanners.
10. Select Add device. Choose the printer when it appears and follow the instructions.
11. If Windows cannot find the printer, install the correct driver from the printer maker's official website.
12. Open the printer's settings, select Print a test page, and check that the page prints correctly.

Result: The printer is installed successfully if it appears in Windows and prints a test page.

Task 3: Fix a printer that is not detected

Problem: The printer is not detected because its USB cable has been unplugged.

Steps:

13. Check that the printer is switched on.
14. Reconnect the USB cable firmly to the printer and the computer.
15. Right-click the Start button and open Device Manager.
16. Look under Printers, Print queues, or Universal Serial Bus controllers for the printer or any device with a warning icon.
17. In Device Manager, select Action > Scan for hardware changes.
18. If the printer appears with an error icon, right-click it and choose Update driver. Follow the instructions.
19. If it still does not appear, try another USB port or a working USB cable, then check Printers & scanners in Windows Settings.

Result: The issue is resolved when Windows detects the printer and it is ready to print.

Task 4: Webcam works in Camera app but not in Zoom

Problem: The webcam works in the Windows Camera app, but the video does not appear in Zoom.

Steps:

20. Close the Camera app and any other app that may be using the webcam.
21. Open Zoom and join or start a test meeting.
22. In Zoom, click the arrow next to Start Video and select the correct webcam from the camera list.
23. Check Windows camera permission: open Settings > Privacy & security > Camera.
24. Turn on Camera access and Let desktop apps access your camera.
25. In Zoom, open Settings > Video and check whether the webcam preview appears.
26. If the video still does not appear, leave Zoom and restart it. If needed, restart the computer and try again.

Result: The problem is fixed when the webcam video appears in Zoom as well as in the Camera app.

Conclusion

In this practical, I learned how to connect and test a USB webcam, install and test a printer, troubleshoot a printer connection, and check camera settings in Zoom. Correct cables, drivers, permissions, and device settings help peripherals work properly.